Manish Shaw

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Indian Institute of Technology (IIT) Bombay | Bachelors in Economics | *2023 – 2027*

Pursuing a **Minor in Artificial Intelligence and Data Science** (C-MInDS)

SCHOLASTIC & PROFESSIONAL ACHIEVEMENTS

- **National Winner**, **Smart India Hackathon** (Govt. of India) among **49K+** competing teams nationwide *(2024)*
- Represented IIT Bombay at the **National Finals**, **Inter-IIT Tech Meet 14.0** against teams from 23 IITs *(2025)*
- Selected as **1 of 4 Claude Campus Ambassadors** in India by **Anthropic** to lead developer adoption *(2026)*
- Selected for the **Y Combinator Startup School** India, a founder programme cohort among **25K+** candidates *(2026)*

PROFESSIONAL EXPERIENCE

AI / Research Engineer | *Otsuka Corporation, Tokyo, Japan* *May'26 – Jul'26*

Operated in a Japanese workplace, delivering technical briefings and benchmarks to stakeholders at Otsuka's Tokyo HQ

- Engineered **AI sales copilot** conditioned on **BOJ Tankan**, modeling consensus for next-best-action at **0.836 AUC**
- Modeled corporate decisions mapping clients across **6 economic regimes**, shipping a **digital-twin** of the boardroom
- Optimized **Mamba-2** + **Nemotron** tool-calling via **LoRA** placement ablations (**42% fewer** params on **BFCL v4**) and **GRPO** post-training, benchmarked using a custom **LLM-as-a-judge** eval harness scoring **recall@k** and **MRR**
- Alignment strategies for tool-calling showed **GRPO** matched **SFT** (**82.8 vs 83.6**) while **DPO** regressed by **5.2 pp**
- Optimized inference for **MoE Qwen3** using **Speculative decoding**, **vLLM** continuous batching, and PagedAttention

Applied AI | *Rebel Foods, Mumbai* *Feb'26 – May'26*

- Deployed an agentic analytics platform routing natural-language queries over **DuckDB** backed by **Amazon Redshift**
- Built **DuckDB** engine over S3 Parquet hydrated by **Apache Airflow**, with partition pruning serving **400+** managers
- Built an LLM **Text-to-SQL** pipeline with **read-only/injection** validation, and golden-set execution benchmarks
- Implemented embedding-based fuzzy entity resolution across **500+** stores using **Voyage AI** vectors to unify analytics

Claude Campus Ambassador | *Anthropic* *Jan'26 – May'26*

- Ran hackathon and workshops on **Claude Code**, **MCP**, **RAG**, multi-agent systems, growing community to **2,000+**
- Owned end-to-end event planning, marketing, onboarding, driving institute-wide adoption of Claude among students

AI & Systems Engineer | *Longani Consulting, Sharjah, UAE* *May'25 – Sep'25*

- Built **BM25+dense** retriever improving **precision@k 35%**; designed CoT/few-shot prompts to reduce hallucinations
- Engineered **FastAPI** with **AsyncIO**, **RBAC**, and **SQLAlchemy ORM** deployed on **Azure** via **BitBucket CI/CD**

KEY PROJECTS

RL Environment for Drift-Robust LLM Agents | *Meta PyTorch OpenEnv Hackathon* *Apr'26*

- **OpenEnv** gym with a Schema-Drift Engine injecting **7** live failure modes on raw wireframes for transfer to real MCP
- Fine-tuned **Qwen2.5-1.5B** via **LoRA/GRPO**, raising valid execution frames from **38%-77%** under live schema drift

Negotiator AI — Multi-Agent Voice Negotiation | *MNLU, Mumbai* *Dec'24 – Feb'26*

- Architected a **6-agent negotiation committee** of specialized role-based agents (Strategist, Legal, Planner, etc.)
- Built a stateful **LangGraph** workflow with checkpointing, mid-flight replanning, HITL interrupts, and **BATNA/ZOPA**
- Implemented low-latency WebRTC/WebSocket voice loop with **STT, VAD**, TTS, **context compaction**, **reflection**
- Grounded agents in **ChromaDB** with cross-encoder reranking; exposed shared session state for agent interoperability

Realtime Credit-Scoring & Fraud Engine | *Inter-IIT Tech Meet 14.0* *Oct'25 – Dec'25*

- Built a Pathway/Kafka streaming fraud engine combining a multi-agent LLM investigator, **FraudGT** graph transformer
- Implemented decision-tree policies for **explainable AI** and orchestrated stateless **Docker** containers on **AWS ECS**

Real-Time Gesture Recognition on Edge Hardware | *SIH - National Winner* *Nov'24 – Dec'24*

- Built a **Seq2Seq** network with Bi-LSTMs and multi-head attention, unifying **MediaPipe**, **Whisper**, and **NeuralMT**
- Quantized PyTorch model to **ONNX INT8**, cutting size by **65%** while sustaining real-time 30 FPS on edge hardware

POSITIONS OF RESPONSIBILITY

Manager, AI Community | *Institute Technical Council, IIT Bombay* *Apr'24 – Present*

- Led **6+ engineers** through Agile sprints, defining production guidelines, unit testing, and deployment for AI systems
- Led **Rizzume** for resume benchmarking/skill-gap scoring and **InstiGPT**, a RAG assistant answering **15K+** queries

TECHNICAL SKILLS

Languages	Python, C++, SQL (PostgreSQL, MySQL), Bash/Shell, Git, Docker, Weights & Biases (WandB)
ML Frameworks	PyTorch (Lightning, DDP, FSDP), TensorFlow, DeepSpeed, TensorRT, Hugging Face, ONNX Runtime, OpenCV, MediaPipe, XGBoost, LightGBM, CatBoost, Scikit-Learn, PyG (GNNs)
LLMs & Inference	LoRA/QLoRA, DoRA, SFT, DPO, GRPO, PPO, RLHF/RLAIF, vLLM, Speculative Decoding, Continuous Batching, PagedAttention, FlashAttention, MoE, AWQ/GPTQ/GGUF Quantization
Agents & RAG	Model Context Protocol (MCP), LangGraph, CrewAI, Multi-Agent Orchestration, Tool/Function Calling, Reflection/Planning Loops, GraphRAG, Hybrid Search, Cross-Encoders, Vector DBs
Data & Cloud	FastAPI, WebSockets/WebRTC, DuckDB, Apache Spark, Kafka, Flink, Airflow, Redis, AWS (SageMaker, ECS, S3), Azure, GCP, Kubernetes (K8s), Terraform, CI/CD, Prometheus, Grafana